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## Smartphone Ownership, Age of Smartphone Acquisition, and Health Outcomes in Early Adolescence

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### Abstract

**Objectives:** Given concerns about health implications of adolescent smartphone use, we tested associations of smartphone ownership and age of smartphone acquisition with depression, obesity, and insufficient sleep in early adolescence. We hypothesized that smartphone ownership, especially at a younger age, would be associated with worse health outcomes.

**Methods:** The sample included 10,588 participants from the Adolescent Brain Cognitive Development Study. Mixed-effects logistic regression models tested associations of smartphone ownership and age of first smartphone acquisition, reported by caregivers, with depression, obesity, and insufficient sleep at age 12. Among participants who did not own smartphones at age 12, we tested associations of recent acquisition of smartphones with outcomes in the subsequent

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Contributors Statement Page

Ran Barzilay contributed to study conceptualization, formulation of research questions and study design, supervision, and wrote the first draft of the manuscript.

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Randy P. Auerbach contributed to study conceptualization and writing the first draft of the manuscript.

All authors substantially contributed to, and approved, the final version of the manuscript.

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year. Models were adjusted for demographic and socioeconomic variables, ownership of other devices, pubertal development, and parental monitoring.

**Results:** At age 12, compared to not owning a smartphone (n=3,849), smartphone ownership (n=6,739) was associated with higher risk for depression (odds ratio [OR], 95% confidence interval [95%CI]=1.31, 1.05–1.63), obesity (OR, 95%CI=1.40, 1.20–1.63), and insufficient sleep (OR, 95%CI=1.62, 1.46–1.79). Younger age of smartphone acquisition was associated with obesity and insufficient sleep (for each earlier year of acquisition, OR, 95%CI=1.09, 1.02–1.16, and 1.08, 1.02–1.12, respectively). At age 13, among 3,486 youth who did not own a smartphone at age 12, those who had acquired a smartphone in the past year (n=1,546) had greater odds of reporting clinical-level psychopathology (OR, 95%CI=1.57, 1.12–2.20) and insufficient sleep (OR, 95%CI=1.50, 1.26–1.77) compared to those who had not (n=1,940), after controlling for baseline mental health and sleep. Results were consistent across several sensitivity analyses.

**Conclusions:** Smartphone ownership was associated with depression, obesity, and insufficient sleep in early adolescence. Findings provide critical and timely insights that should inform caregivers regarding adolescent smartphone use and ideally, the development of public policy that protects youth.

## INTRODUCTION

Smartphone ownership is a *rite of passage* for youth, and in the United States, approximately half of children will have their own smartphone by age 11.<sup>1</sup> Smartphones provide ready and immediate access to schoolwork, social networks, internet browsing, gaming, and social media. Given the potential for negative exposures (e.g., bullying, explicit content) coupled with the fact that many 8–12-years-olds are spending ~4–6 hours per day using screens, inclusive of smartphones,<sup>2</sup> and potential adverse health associations of screen time,<sup>3,4</sup> there is growing concern about how smartphone use affects mental and physical health among young people. Accordingly, the American Academy of Pediatrics (AAP) and the American Academy of Child and Adolescent Psychiatry (AACAP) have established guidelines to support families in instituting healthier boundaries, delineating recommended screen time limits, preferred parental controls, and optimal parental modeling practices.<sup>5</sup> To our knowledge, however, there are no guidelines regarding the appropriate age at which youth should first own a smartphone.<sup>6,7</sup>

The decision to provide a child with their own smartphone is often initiated as a matter of necessity, allowing communication with caregivers and addressing safety concerns. Although the intended use for smartphones may be circumscribed, there is a natural drift wherein youth begin to use their smartphones for other activities (e.g., socialization, education).<sup>6</sup> Most prior research has focused on the consequences of excessive, addictive or problematic smartphone use,<sup>8–12</sup> reflecting a preoccupation and relative loss of control, and reported among 3%–35% of youth.<sup>13</sup> Problematic smartphone use is associated with a wide range of negative outcomes, including adverse mental health consequences (e.g., depression), increased sedentary behaviors, and diminished sleep.<sup>9,14,15</sup> However, it is not evident whether simply owning a smartphone, accounting for the age of ownership, in childhood and early adolescence potentiates risks for negative health consequences.

As smartphone ownership among children becomes more ubiquitous, there is a critical need for data to inform families regarding the decision about when to provide their child access to smartphones and to guide public health policy. Thus, this study aimed to examine how acquiring a smartphone relates to health outcomes in early adolescence. Accordingly, we used data from the Adolescent Brain Cognitive Development (ABCD) Study which tracks US youth annually from late childhood. We hypothesized that smartphone ownership and earlier acquisition of smartphones would be associated with depression, obesity, and insufficient sleep.

## METHODS

### PARTICIPANTS

ABCD Study participants were enrolled across 21 sites throughout the US and assessed from late childhood into adolescence.<sup>16</sup> Therefore, the baseline assessment was completed at mean age ~10, with follow-up assessments each subsequent year (e.g., 1-year follow-up at mean age ~11, 2-year follow-up at age ~12, 3-year follow-up at age ~13). The ABCD study conducts in-person structured clinical computerized assessments, while BMI is measured by taking height and weight of youth study participants. ABCD data release 5.1, which included five assessments conducted between 2016–2022 (baseline to 4-year-follow-up, age range 9–16), was utilized. Smartphone ownership was first assessed in the 2-year follow-up assessment, and thus, we used data from this assessment and the subsequent assessments. Sample description is summarized in Supplemental Table S1. Note that data on less than half of the sample was available for the 4-year follow-up assessment. This data was used only in supplemental sensitivity analyses. The ABCD Study protocol was approved by the University of California, San Diego Institutional Review Board (IRB), and analyses were exempted from review by the Children's Hospital of Philadelphia IRB. We adhered to the “strengthening the reporting of observational studies in epidemiology (STROBE)” guidelines.

### SMARTPHONE OWNERSHIP

Smartphone ownership was assessed annually starting at the 2-year follow-up assessment based on caregiver response to the question, “*Does your child have their own cellphone?*” and to the follow-up question regarding device type. Participants with an iPhone or Android were considered as owning a smartphone. Age of receiving a smartphone was identified based on caregiver response to the question, “*How old was your child when he/she got their own cellphone?*” First-time smartphone ownership in the past year was considered for youth whose caregiver responded “no” regarding cellphone ownership at the 2-year follow-up but later responded “yes” at the 3-year follow-up.

### HEALTH-RELATED OUTCOMES

To enhance clinical interpretability and effect size interpretation, we used binary outcomes in our main analyses. Mental health outcomes included lifetime-depression diagnosis based on the “Kiddie Schedule for Affective Disorders and Schizophrenia Child Psychiatric Diagnostic Interview” (KSADS<sup>17</sup>) at the 2-year and 4-year follow-up assessments; and clinical threshold level (T-score  $\geq 65$ <sup>18</sup>) of psychopathology at the 3-year follow-up (that did

not include KSADS assessment) based on the self-reported Brief Problem Monitor (BPM<sup>19</sup>) scale, which assesses internalizing, externalizing, and attention symptoms.<sup>18</sup> Obesity was determined by body mass index (BMI, kg/m<sup>2</sup>, based on weight/height measurements during study assessment) over 95<sup>th</sup> percentile.<sup>20</sup> Insufficient sleep was defined as youth-reported sleep duration of less than 9 hours per night.<sup>21</sup> Supplemental eMethods provide further details on outcome measures.

## STATISTICAL ANALYSES

We conducted three sets of analyses to test our hypotheses. Analysis 1 tested associations of smartphone ownership with outcomes at age 12 (2-year follow-up). Analysis 2 examined how the age at which participants first acquired a smartphone (in years; continuous measure), was associated with health outcomes at age 12, focusing specifically on the subgroup who already owned a smartphone by that age. Analysis 3 investigated the associations between first acquiring a smartphone in the past year and outcomes at age 13. Analysis 3 included only participants without smartphones at age 12, creating a cohort that compares those who subsequently acquired smartphones versus those who remained without smartphones by age 13, ensuring both groups shared a common pre-exposure period. This allows pre-exposure outcome measures, assessed at age 12 (psychopathology, obesity, insufficient sleep), to be adjusted for as covariates, providing a higher quality of causal evidence.

Mixed-effects logistic regression models (“mixed models”) used smartphone ownership (Analysis 1), age of first smartphone acquisition (Analysis 2), or past-year smartphone acquisition (Analysis 3) as independent variables, with depression, obesity, and insufficient sleep as outcome variables. Analysis 3 used clinical-threshold psychopathology as the outcome instead of depression as the KSADS was not used at the 3-year follow-up assessment. Mixed models accounted for the hierarchical nature of the data using nested random effects for family (when siblings were enrolled) and research site. To correct for the testing of three outcomes, we applied a P-value threshold of 0.017 (0.05/3). For each outcome, we estimated a base model that included age and sex assigned at birth, and an adjusted model that co-varied for confounders including household income, parent education, participants’ race and ethnicity (caregiver-report), pubertal status (pre-, early, mid-, late, and post pubertal<sup>22</sup>), and parental monitoring (average of five youth-reported items,<sup>23</sup> each rated from 1 (Never) to 5 (Always)), which is associated with the studied outcomes.<sup>24–26</sup> To enhance specificity of results to smartphone ownership, this model adjusted for ownership of other devices (tablet, laptop, iPod, and smartwatch). Covariates are described in eMethods. Note that the rationale to include race and ethnicity as covariates<sup>27</sup> was to address potential confounding due to differences in smartphone ownership (exposure variable) and health outcomes among racial and ethnic groups in ABCD.<sup>28,29</sup> We explored sex differences by adding the interaction term (product) of the exposure and sex to the adjusted model. Models utilized listwise deletion of missing data. Data missingness is provided in Tables 1–2. As no analysis included multiple timepoints with smartphone ownership, adjustments for dropout and intention-to-treat analyses were not required. Comparison of participants included versus excluded due to missing exposure data (<10%) is provided in Supplemental Table S2.

For each model, we report the odds ratio (OR), exponentiated regression coefficient, 95% confidence interval (95%CI), and two-sided P-value. In Analysis 2, coefficients were transformed to reflect a positive relationship between younger age and adverse outcome (depression, obesity, insufficient sleep) by using the inverse of the OR and 95%CI. We used R (version 4.5.0) package lme4 for statistical models and ggplot package for data visualization.

## SENSITIVITY ANALYSES

1. We re-estimated models excluding participants with problematic smartphone use. As there is no gold-standard tool to quantify problematic smartphone use,<sup>30</sup> we used the Mobile Phone Involvement Questionnaire<sup>31</sup> that probes problematic smartphone use patterns in ABCD<sup>32</sup> (detailed in eMethods).
2. As our analysis of observational data depends on an assumption of no unobserved confounding between exposed versus unexposed conditions, we conducted E-value sensitivity analysis<sup>33</sup> to quantify the degree of unobserved confounding required to explain our results in the absence of a treatment effect (detailed in eMethods).
3. We re-estimated models with continuous outcome measures (overall psychopathology, BMI, hours of sleep) using linear mixed models.
4. We re-estimated mixed models of Analyses 1–2 using longitudinal data from the 2-year-, 3-year-, and 4-year-follow-up assessments.

## RESULTS

### SMARTPHONE OWNERSHIP

Among youth who reported owning a cellphone, nearly all (6,739 of 7,010; 96.1%) owned a smartphone (iPhone or Android, Table 1). Smartphone ownership rose steadily across the assessments: 63.6% (6,739/10,588) at 2-year-follow-up (mean age 12.02), 77.2% (7,602/9,853) at 3-year-follow-up (mean age 12.91), and 89.1% (4,094/4,593) at 4-year-follow-up (mean age 14.08). The median age of first smartphone acquisition was 11 years old (range=4–13, interquartile range 10–11) (Figure 1).

### ANALYSIS 1: SMARTPHONE OWNERSHIP AND HEALTH OUTCOMES AT AGE 12

Compared with their peers who did not own smartphones, participants who owned smartphones tended to be slightly older, at a more advanced pubertal stage, more likely to be female, more likely to identify as Black or of Hispanic ethnicity, and more likely to be from households with lower income and lower parental educational attainment (Table 1). Participants with smartphones reported more depression, obesity, and insufficient sleep (Table 1). Correlations among psychopathology, BMI, and sleep were small (Pearson  $|r|$  values  $\leq 0.20$ , all  $P$ 's  $< 0.001$ ; Supplemental Figure S1).

In adjusted models (Figure 2A; Supplemental Tables S3–5), owning a smartphone was associated with depression (OR, 95%CI=1.31, 1.05–1.63,  $P=0.016$ ), obesity (OR,

95%CI=1.40, 1.20–1.63,  $P<.001$ ), and insufficient sleep (OR, 95%CI=1.62, 1.46–1.79,  $P<.001$ ).

## **ANALYSIS 2: AGE OF FIRST SMARTPHONE ACQUISITION AND HEALTH OUTCOMES AT AGE 12**

In adjusted models (Figure 2B; Supplemental Tables S6–8), younger age of first smartphone acquisition was associated with obesity (for each earlier year of acquisition, OR, 95%CI=1.09, 1.02–1.16,  $P=0.007$ ) and insufficient sleep (OR, 95%CI=1.08, 1.02–1.12,  $P=0.005$ ), but not with depression (OR, 95%CI=1.03, 0.94–1.12,  $P=0.503$ ). Note that results from the base model, which only co-varied for age and sex, showed an association of earlier age of smartphone ownership with depression, but this did not satisfy the  $P$ -value threshold (OR, 95%CI=1.08, 1.0–1.15,  $P=0.046$ ).

## **ANALYSIS 3: RECENT (PAST YEAR) ACQUISITION OF A SMARTPHONE AND HEALTH OUTCOMES AT AGE 13**

We analyzed data from 3,486 youth who did not own smartphones at the 2-year follow-up and who also participated in the 3-year follow-up, when 1,546 had acquired a smartphone and 1,940 had not. Table 2 details comparison of the groups. In adjusted models (Supplemental Tables S9–11), youth who acquired a smartphone by the 3-year follow-up were more likely to reach clinical levels of psychopathology (OR, 95%CI=1.57, 1.12–2.20,  $P=0.009$ ; Figure 3) and to sleep less than 9 hours per night (OR, 95%CI=1.50, 1.26–1.77,  $P<0.001$ ; Figure 3), with no statistically significant difference in obesity (OR, 95%CI=1.57, 0.82–3.04,  $P=0.176$ ; Figure 3), compared to those who did not.

## **SEX DIFFERENCES**

We did not detect significant interactions with sex in any of the smartphone exposures.

## **SENSITIVITY ANALYSES**

Analyses excluding participants who engage in problematic smartphone use yielded similar results to main analyses in direction and in statistical significance in all adjusted models (Supplemental Table S12). E-value analyses addressing unobserved confounding run on fully adjusted models suggested that implausibly strong unmeasured confounders would be required to explain away the observed associations (Supplemental Table S13). Analyses that used continuous outcomes (overall psychopathology BPM score, BMI, hours of sleep, Supplemental Tables S14–22), and analyses that included longitudinal data from the 2-year, 3-year and the 4-year follow-up assessment waves (Supplemental Figure S2, Supplemental Tables S23–28), revealed similar results in direction and statistical significance as main analyses, except for smartphone by sex interaction that became significant in association with sleep (Supplemental Table 25; interaction  $P=0.001$ ), with girls showing stronger associations between smartphone ownership and insufficient sleep than boys.

## **DISCUSSION**

In this large-scale multi-site study of American youth of early-adolescent age, we provide converging evidence for adverse associations of smartphone ownership with

health outcomes, accounting for several demographic, socioeconomic, and parental factors. Smartphone ownership at age 12 was related to depression, obesity, and insufficient sleep (Analysis 1). Earlier age of smartphone acquisition was associated with obesity and insufficient sleep at age 12 (Analysis 2). Recent (past-year) first-time acquisition of smartphones by age 13 was associated with clinical-threshold psychopathology and insufficient sleep, even when accounting for psychopathology and sleep in the prior year (Analysis 3). Results appear specific to smartphones as our models accounted for ownership of other devices including tablets, laptops, iPods, and smartwatches. These findings contribute to a growing body of research indicating that an earlier age of smartphone ownership may be linked to poorer mental health outcomes. Prior studies have reported that early smartphone acquisition is associated with greater psychological distress among Spanish teens,<sup>34</sup> and that receiving a smartphone before age 13 is associated with worse social and emotional health in young adulthood compared to those who received a smartphone after age 13 in a multi-national sample.<sup>35</sup> Notably, the only US study on this topic focused on low socioeconomic status Hispanic youth with overweight or obesity and found no adverse association between smartphone ownership and either mental health or sleep.<sup>36</sup> This contrast highlights that findings from a narrow population may not apply broadly, strengthening the generalizability of our findings that were derived from a diverse US cohort from a multi-site study.

No clear evidence exists on the explanation for the deleterious associations between smartphone ownership and youth development. However, there is evidence that smartphone use contributes to fragmented attention, increased checking behaviors (which may impact relationships), and extended use (particularly in the evenings), which may lead to both mental and physical health challenges and less sleep,<sup>9,10</sup> especially during a period of development whereby many youth may not have sufficiently mature self-regulatory skills to make optimal choices regarding their smartphone use.<sup>37</sup> Importantly, our results are separable from problematic smartphone use, as our sensitivity analysis that excluded youth with problematic smartphone use suggest that smartphone ownership, even when not rising to the level of problematic use, may negatively impact adolescent mental and physical health, which has substantial public health implications.

A key finding is that an earlier age of first smartphone acquisition was associated with higher rates of obesity and insufficient sleep at age 12 in models that adjusted for age, sex, and multiple confounders. Links to higher rates of depression when adjusting for age and sex ( $P=0.046$ ) did not satisfy our  $P$ -threshold. We highlight the developmental context of these results. Prior literature focused on addictive or excessive smartphone use suggests that smartphone ownership around childhood or early adolescent age may displace time spent on in-person interactions,<sup>38</sup> physical activity,<sup>39</sup> and restorative sleep,<sup>40</sup> key foundations for adolescent health. Notably, although the effect sizes for the age of first smartphone acquisition are small (with odds ratios for all three outcomes around 1.1 for each earlier year of ownership), adolescence is a sensitive period during which even modest subclinical mental health symptoms, overweight, and sleep problems can have long-term health consequences into adulthood.<sup>41–43</sup> Thus, small effect sizes at the individual level may translate to substantial public health significance. Another important developmental consideration involves sex differences during this age range, which coincides

with the transition to puberty. Previous research in older adolescents suggests that girls may experience stronger adverse effects of smartphone use, particularly in relation to mental health outcomes such as depression.<sup>14</sup> However, our exploratory analyses did not detect significant smartphone ownership by sex differences in association with depression at early adolescent age. It is possible that such differences emerge later in adolescence, coinciding with the increase in depression rates among girls following the onset of puberty.<sup>44</sup> Note that our sensitivity analysis that leveraged data from multiple timepoints, increasing the number of observations, did reveal a significant smartphone by sex interaction in association with impaired sleep, suggesting greater susceptibility among girls. Future longitudinal work from early- to late-adolescence should re-examine sex differences in the ways teens use their smartphones and how this relates to health outcomes.

Given our findings, it is evident that a concrete framework is needed to advise on childhood and early adolescent smartphone ownership to support the healthier development of youth. There are existing tools available to gauge youth readiness. For example, AAP has developed an easily accessible questionnaire, the *Phone Ready Questionnaire*,<sup>45</sup> which is a thoughtful instrument addressing a range of key youth (e.g., need, judgement, impulsiveness, accountability) and parental factors (e.g., readiness to impose boundaries) to consider prior to determining whether a child is ready to have a smartphone. Although AAP and AACAP offer other *common-sense* guidelines for instituting safer practices, empirical research is needed to establish clearer guidelines specific to: (a) optimal age of ownership, (b) type of phone (e.g., starter phone with more limited capabilities), and (c) uses (e.g., limiting to texts, phone calls, restricting access to specific apps).

Study strengths include a large longitudinal cohort from diverse communities, but there are also notable limitations. The focus of this study was on smartphone ownership, but it did not provide insight into *what* youth were doing on their smartphones. Factors such as patterns of use and exposure to adverse content may contribute to health outcomes in ways distinct from age. Additionally, although our findings largely persisted when accounting for parental monitoring, there may be other important, unexplored factors that buffer against negative health outcomes. As this study used observational data (a randomized placebo-controlled trial would not have been possible), we cannot be certain that unmeasured factors are not already more prevalent among youth without smartphones, prompting caution in drawing causal conclusions from our findings. We address the research question within these limitations via converging analyses, careful inclusion of key confounders, and sensitivity analysis for unobserved confounding.

Although our findings may signal a need for public policy changes, we are not advocating for restrictions on youth access and ownership of smartphones. Rather, we highlight the need for further research to better understand the adverse effects of smartphone ownership during the transition from childhood to adolescence, a developmental period that is already high-risk for psychiatric disorders<sup>46,47</sup> and sleep problems.<sup>43,48</sup> More broadly, smartphone ownership offers unique challenges as it may grant youth unfettered access to a world for which they may not be ready, without the discipline to effectively manage their own use. Like youth's diet, media consumption (e.g., television, social media), and relationships (e.g., with peers, adults), smartphone ownership requires dedicated oversight. This is particularly

true because there may be various adverse consequences and challenges for certain youth who *do not* own smartphones, which underscores the need to protect these youth who require smartphones and support families as they embark on this *rite of passage*.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development<sup>SM</sup> (ABCD) Study (<https://abcdstudy.org>), held in the NIMH Data Archive (NDA). This is a multisite, longitudinal study designed to recruit more than 10,000 children age 9–10 and follow them over 10 years into early adulthood. The ABCD Study<sup>®</sup> is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at [https://abcdstudy.org/consortium\\_members/](https://abcdstudy.org/consortium_members/). ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or ABCD consortium investigators. The authors thank Christina Ly for supporting the literature review for this project.

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Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive Development Study (<https://abcdstudy.org>), held in the National Institute of Mental Health Data Archive and available for researchers upon application (<https://nda.nih.gov/abcd/request-access>). Analytic code describing all analyses is available at [https://github.com/barzilab1/ABCD\\_smartphone\\_health](https://github.com/barzilab1/ABCD_smartphone_health).

### Abbreviations:

|                   |                                              |
|-------------------|----------------------------------------------|
| <b>ABCD Study</b> | Adolescent Brain Cognitive Development Study |
| <b>BMI</b>        | body mass index                              |
| <b>BPM</b>        | brief problem Monitor                        |

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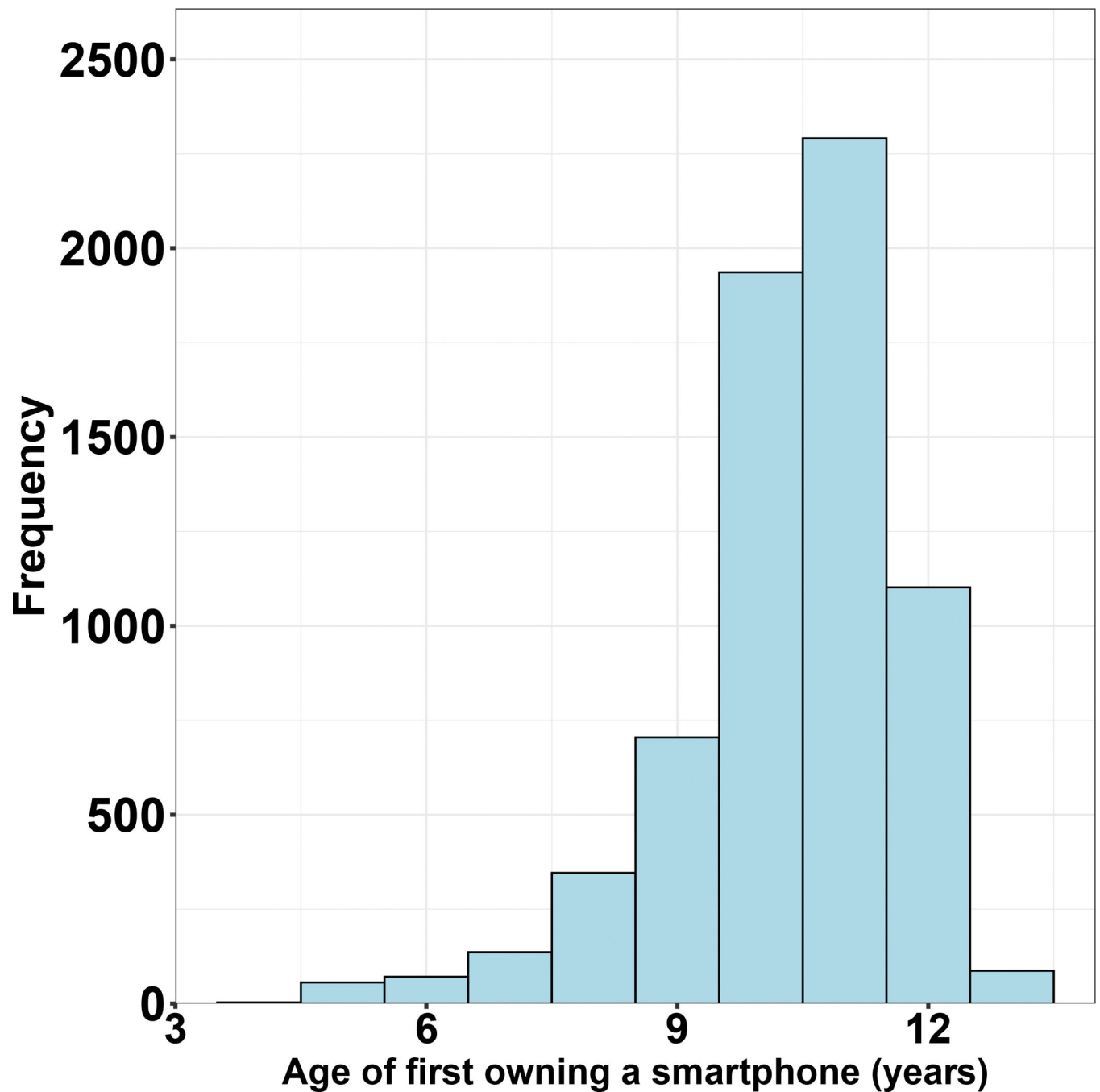
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**What's Known on This Subject**

Most American youth acquire a smartphone during early adolescence. Previous research has linked general screen use around this age, but not specifically smartphone use, to impairments in mental health, body weight, and sleep patterns.

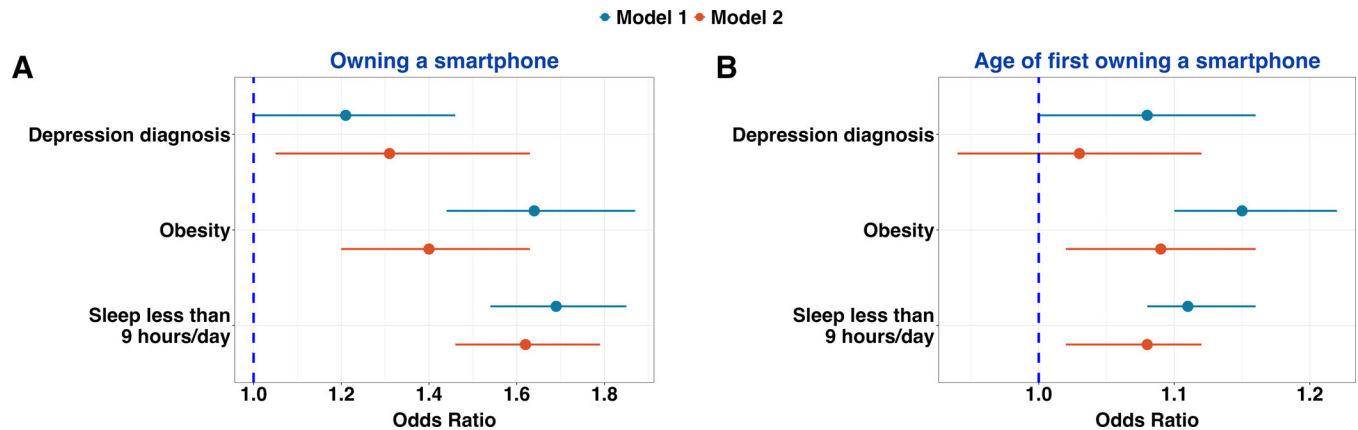
**What This Study Adds**

Among US youth of early adolescent age, smartphone ownership and earlier age of smartphone acquisition is associated with higher rates of depression, obesity, and insufficient sleep. These findings highlight the need to consider health risks when teens gain smartphone access.



**Figure 1: Distribution of age of smartphone acquisition.**

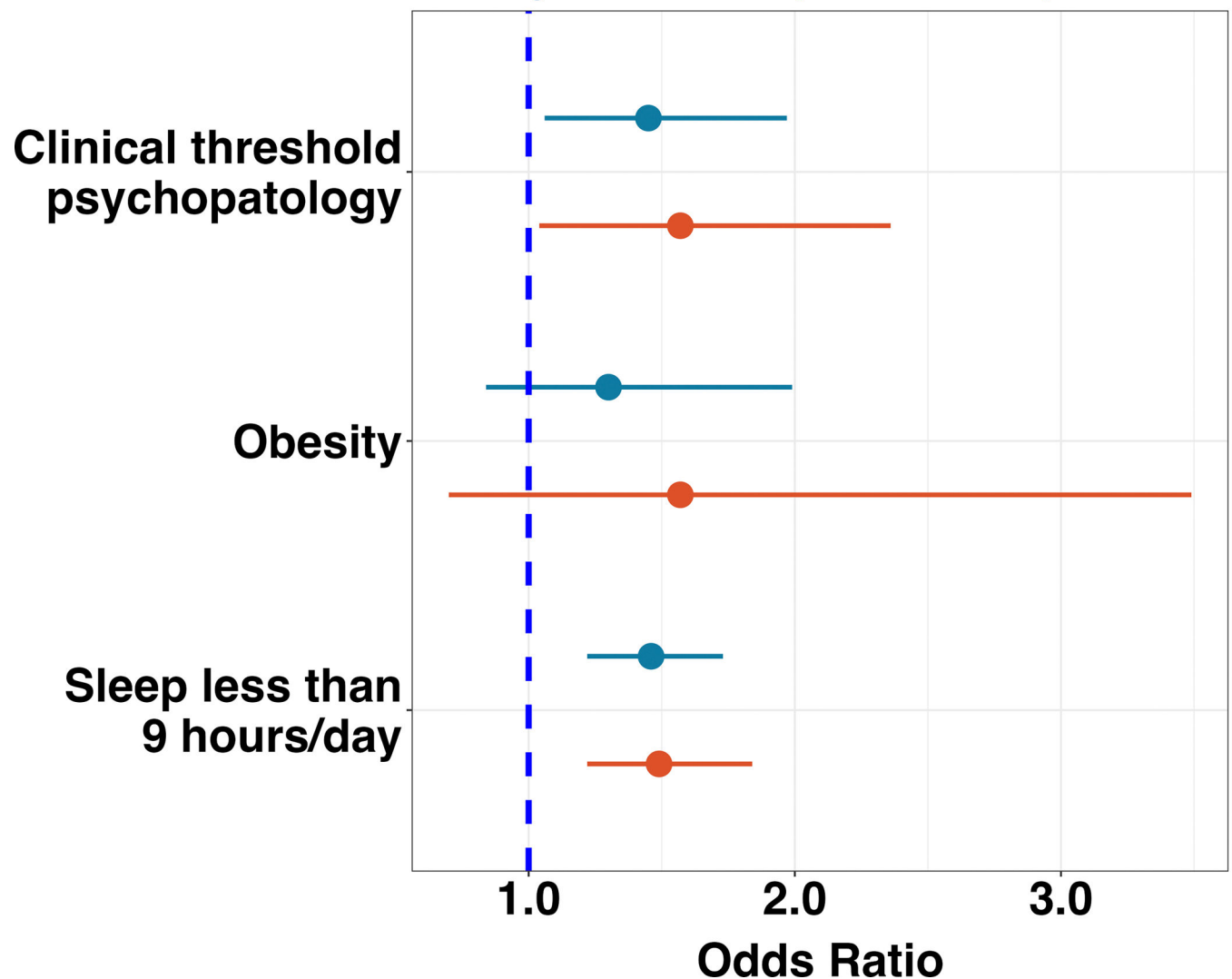
Histogram depicting the distribution of age at which ABCD Study participants first owned a smartphone as reported at the 2-year follow-up ABCD assessment wave. Abbreviation: ABCD, Adolescent Brain Cognitive Development.



**Figure 2: Association of smartphone ownership and age of first smartphone ownership as reported at the 2-year follow-up ABCD assessment wave with health outcomes at age 12.**

**(A)** Association of smartphone ownership and **(B)** age of first smartphone acquisition with depression, obesity, and insufficient sleep as captured at the 2-year follow-up assessment wave of ABCD Study (age 12 years). Model 1 adjusted for age and sex. Model 2 further adjusted for race, ethnicity, income, parent education, ownership of other devices (eg, tablet, laptop, iPod, smartwatch), pubertal status, and parental monitoring. Dots represent OR and 95% CI values. Vertical dashed lines represent the null hypothesis for no association between smartphone ownership and outcome. Note that, for visualization purposes, coefficients in panel B were transformed to reflect a positive relationship between younger age and adverse outcome (more depression, more obesity, insufficient sleep) by using the inverse of the OR and 95% CI values. Abbreviations: ABCD, Adolescent Brain Cognitive Development; OR, odds ratio.

## Past-year smartphone acquisition



**Figure 3: Association of first acquiring a smartphone in the past year with health outcomes in the 3-year follow-up ABCD Study assessment wave (age 13).**

Association of first smartphone acquisition over the past-year with clinical-threshold psychopathology, obesity, and insufficient sleep as captured at the 3-year follow-up assessment wave of ABCD Study (age 13 years). Model 1 adjusted for age and sex. Model 2 further adjusted for race, ethnicity, income, parent education, ownership of other devices (eg, tablet, laptop, iPod, smartwatch), pubertal status, parental monitoring, and the measurement of the index outcome a year earlier, ie, at the 2-year follow-up assessment. Psychopathology was measured with the self-reported BPM (age and sex standardized T-score), clinical threshold was considered for BPM score greater than or equal to 65. Dots represent OR and 95% CI values. Vertical dashed lines represent the null hypothesis for no association between past-year smartphone acquisition and outcome. Abbreviations: ABCD, Adolescent Brain Cognitive Development; BPM, brief problem monitor; OR, odds ratio.

**Table 1.**

Comparison of youth who own a smartphone to those who do not at the 2-year follow-up assessment wave.

|                                                     | Overall sample | Does not own a smartphone | Owns a smartphone | Uncorrected P-value | Missing data, % |
|-----------------------------------------------------|----------------|---------------------------|-------------------|---------------------|-----------------|
| n                                                   | 10,588         | 3,849                     | 6,739             |                     |                 |
| Smartphone type, iPhone                             |                | NA                        | 4,313             |                     |                 |
| Smartphone type, Android                            |                | NA                        | 2,426             |                     |                 |
| <b>Demographic characteristics</b>                  |                |                           |                   |                     |                 |
| Age at 2-year F/U, mean (SD)                        | 12.02 (0.67)   | 11.83 (0.63)              | 12.13 (0.66)      | <0.001              | 0.01%           |
| Late/post pubertal (2-year F/U), n (%)              | 2,122 (20.04)  | 480 (12.47)               | 1,642 (24.37)     | <0.001              | 5.37%           |
| Female sex assigned at birth, n (%)                 | 5,028 (47.49)  | 1,623 (42.17)             | 3,405 (50.53)     | <0.001              | 0.03%           |
| White race, n (%)                                   | 7,985 (75.42)  | 3,161 (82.13)             | 4,824 (71.58)     | <0.001              | 0.17%           |
| Black race, n (%)                                   | 2,153 (20.33)  | 492 (12.78)               | 1,661 (24.65)     | <0.001              | 0.17%           |
| Asian, n (%)                                        | 668 (6.31)     | 272 (7.07)                | 396 (5.88)        | 0.02                | 0.17%           |
| American Indian and Alaska Native, n (%)            | 365 (3.45)     | 97 (2.52)                 | 268 (3.98)        | <0.001              | 0.17%           |
| Mixed race, n (%)                                   | 1,282 (12.11)  | 413 (10.73)               | 869 (12.90)       | 0.001               | 0.17%           |
| Hispanic ethnicity, n (%)                           | 2,067 (19.52)  | 621 (16.13)               | 1,446 (21.46)     | <0.001              | 1.17%           |
| Household income, <sup>1</sup> mean (SD)            | 7.54 (2.31)    | 7.86 (2.11)               | 7.35 (2.40)       | <0.001              | 8.03%           |
| Parent highest education, <sup>2</sup> n (%)        |                |                           |                   | <0.001              | 0.19%           |
| Highschool or below                                 | 449 (4.24)     | 135 (3.51)                | 314 (4.66)        |                     |                 |
| Highschool graduate                                 | 976 (9.22)     | 229 (5.95)                | 747 (11.08)       |                     |                 |
| Post high school education                          | 2,571 (24.28)  | 713 (18.52)               | 1,858 (27.57)     |                     |                 |
| Bachelor                                            | 2,738 (25.86)  | 1,060 (27.54)             | 1,678 (24.90)     |                     |                 |
| Master or above                                     | 3,834 (36.21)  | 1,708 (44.38)             | 2,126 (31.55)     |                     |                 |
| <b>Ownership of other devices</b>                   |                |                           |                   |                     |                 |
| Tablet, n (%)                                       | 4,696 (44.35)  | 1,896 (49.26)             | 2,800 (41.55)     | <0.001              | 0%              |
| Laptop, n (%)                                       | 4,026 (38.02)  | 1,009 (26.21)             | 3,017 (44.77)     | <0.001              | 0%              |
| iPod, n (%)                                         | 1,144 (10.80)  | 719 (18.68)               | 425 (6.31)        | <0.001              | 0%              |
| Smartwatch, n (%)                                   | 619 (5.85)     | 109 (2.83)                | 510 (7.57)        | <0.001              | 0%              |
| <b>Clinical characteristics at 2-year follow-up</b> |                |                           |                   |                     |                 |
| Psychopathology symptoms (mean [SD])                | 53.71 (5.64)   | 53.35 (5.46)              | 53.91 (5.73)      | <0.001              | 11.46%          |
| Clinical threshold psychopathology, n (%)           | 906 (8.56)     | 289 (7.51)                | 617 (9.16)        | <0.001              | 11.46%          |
| Lifetime depression diagnosis, n (%)                | 604 (5.70)     | 174 (4.52)                | 430 (6.38)        | <0.001              | 0.71%           |
| BMI, kg/m <sup>2</sup> (mean [SD])                  | 20.75 (4.99)   | 19.84 (4.43)              | 21.30 (5.21)      | <0.001              | 16.10           |
| Obesity, n (%)                                      | 1649 (15.57)   | 454 (11.80)               | 1195 (17.73)      | <0.001              | 16.13%          |
| Sleep duration, hours (mean [SD])                   | 9.12 (1.26)    | 9.40 (1.15)               | 8.97 (1.29)       | <0.001              | 0.96%           |
| Insufficient sleep (<9h), n (%)                     | 4,372 (41.29)  | 1,194 (31.02)             | 3,178 (47.16)     | <0.001              | 0.96%           |

Univariate comparisons were conducted for descriptive purposes to compare youth who did versus did not own a smartphone: t-test for normally distributed continuous variables, Mann-Whitney U test for non-normal continuous variables, and chi-square test for categorical variables. If the expected frequencies in any cell of the contingency table were less than 5, we used Fisher's exact tests. Abbreviations: F/U, follow-up; SD, standard deviation; BMI, body mass index.

<sup>1</sup>Income is expressed as the mean of the values from ten self-reported income brackets:(1) <\$5,000; (2) \$5,000–12,000; (3) \$12,000–16,000; (4) \$16,000–25,000; (5) \$25,000–35,000; (6) \$35,000–50,000; (7) \$50,000–75,000; (8) \$75,000–100,000; (9) \$100,000–200,000; (10) >\$200,000.

<sup>2</sup>Parental education represents the highest level of education attained by either parent.

**Table 2.**

Comparison of youth without smartphones at 2-year follow-up: Those who acquired a smartphone by 3-year follow-up vs. those who did not

|                                                     | Overall sample | No smartphone at follow-up | Owens a smartphone at follow-up | Uncorrected P-value | Missing data, % |
|-----------------------------------------------------|----------------|----------------------------|---------------------------------|---------------------|-----------------|
| n                                                   | 3,486          | 1,940                      | 1,546                           |                     |                 |
| Smartphone type, iPhone                             |                | NA                         | 1,017                           |                     |                 |
| Smartphone type, Android                            |                | NA                         | 529                             |                     |                 |
| <b>Demographic characteristics</b>                  |                |                            |                                 |                     |                 |
| Age at 2-year F/U, mean (SD)                        | 11.82 (0.63)   | 11.79 (0.62)               | 11.85 (0.64)                    | 0.006               | 0.03%           |
| Late/post pubertal (2-year F/U), n (%)              | 416 (11.93)    | 203 (10.46)                | 213 (13.78)                     | 0.004               | 4.96%           |
| Female sex assigned at birth, n (%)                 | 1,458 (41.82)  | 775 (39.95)                | 683 (44.18)                     | 0.01                | 0.03%           |
| White race, n (%)                                   | 2897 (83.10)   | 1640 (84.54)               | 1257 (81.31)                    | 0.02                | 0.2%            |
| Black race, n (%)                                   | 421 (12.08)    | 200 (10.31)                | 221 (14.29)                     | <0.001              | 0.2%            |
| Asian, n (%)                                        | 252 (7.23)     | 164 (8.45)                 | 88 (5.69)                       | 0.002               | 0.2%            |
| American Indian and Alaska Native, n (%)            | 94 (2.70)      | 53 (2.73)                  | 41 (2.65)                       | 0.98                | 0.2%            |
| Mixed race, n (%)                                   | 383 (10.99)    | 220 (11.34)                | 163 (10.54)                     | 0.51                | 0.2%            |
| Hispanic ethnicity, n (%)                           | 552 (15.83)    | 276 (14.23)                | 276 (17.85)                     | 0.004               | 1.49%           |
| Household income, <sup>1</sup> mean (SD)            | 7.94 (2.09)    | 8.00 (2.00)                | 7.87 (2.19)                     | 0.09                | 6.63%           |
| Parent highest education, <sup>2</sup> n (%)        |                |                            |                                 | <0.001              | 0.2%            |
| Highschool or below                                 | 111 (3.18)     | 52 (2.68)                  | 59 (3.82)                       |                     |                 |
| Highschool graduate                                 | 187 (5.36)     | 87 (4.48)                  | 100 (6.47)                      |                     |                 |
| Post high school education                          | 614 (17.61)    | 299 (15.41)                | 315 (20.38)                     |                     |                 |
| Bachelor                                            | 985 (28.26)    | 554 (28.56)                | 431 (27.88)                     |                     |                 |
| Master or above                                     | 1,582 (45.38)  | 941 (48.51)                | 641 (41.46)                     |                     |                 |
| <b>Clinical characteristics at 2-year follow-up</b> |                |                            |                                 |                     |                 |
| Psychopathology symptoms (mean [SD])                | 53.31 (5.46)   | 53.17 (5.38)               | 53.48 (5.57)                    | 0.11                | 11.42%          |
| Clinical threshold psychopathology, n (%)           | 266 (7.63)     | 140 (7.22)                 | 126 (8.15)                      | 0.44                | 11.42%          |
| Lifetime depression diagnosis, n (%)                | 150 (4.30)     | 86 (4.43)                  | 64 (4.14)                       | 0.73                | 0.72%           |
| BMI, kg/m <sup>2</sup> (mean [SD])                  | 19.76 (4.34)   | 19.44 (4.07)               | 20.14 (4.62)                    | <0.001              | 12.79%          |
| Obesity, n (%)                                      | 408 (11.70)    | 198 (10.21)                | 210 (13.58)                     | 0.005               | 12.82%          |
| Sleep duration, hours (mean [SD])                   | 9.40 (1.14)    | 9.47 (1.12)                | 9.30 (1.15)                     | <0.001              | 0.86%           |

Univariate comparisons were conducted for descriptive purposes: t-test for normally distributed continuous variables, Mann-Whitney U test for non-normal continuous variables, and chi-square test for binary measures. If the expected frequencies in any cell of the contingency table were less than 5, we used Fisher's exact tests. Abbreviations: F/U, follow-up; SD, standard deviation; BMI, body mass index.

<sup>1</sup>Income is expressed as the mean of the values from ten self-reported income brackets: (1) <\$5,000; (2) \$5,000–12,000; (3) \$12,000–16,000; (4) \$16,000–25,000; (5) \$25,000–35,000; (6) \$35,000–50,000; (7) \$50,000–75,000; (8) \$75,000–100,000; (9) \$100,000–200,000; (10) >\$200,000.

<sup>2</sup>Parental education represents the highest level of education attained by either parent.